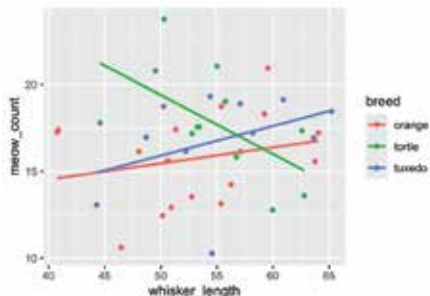


Anatomy of a ggplot



```
# initialise with ggplot()
ggplot(data = cat_data,
# choose aesthetics for
# variables in your data
aes(x = whisker_length,
y = meow_count,
colour = breed)) +
# add geoms (shapes) that
# physically depict data
geom_point() +
# keep adding layers with +
geom_smooth(method = "lm")
```

Commonly used geoms

geom_boxplot() 

geom_violin() 

geom_point() 

geom_bar() 

geom_histogram() 

geom_hline()/geom_vline() 

geom_smooth() 

Saving plots

path of file to save to
 ↓
 ggsave("my_plot.png",
 plot = my_plot, height = 5,
 width = 5) ↑
 variable containing your plot

Colour scales

- For **categorical data**, use `scale_colour_manual(values = c("red", "pink", "orange"))`
- For **continuous data**, use `scale_colour_gradient(high = "red", low = "red")`
- Consider trying pre-made palettes e.g. `viridis`) or specifying hex colours (e.g. `"#f932ec"`)

Scales for the x and y axes

- set limits with `xlim(min, max)` or `ylim()`
- apply transformations with e.g. `scale_x_log10()`
- edit labelling using `labels` argument to any `scale_*` function

Position adjustments

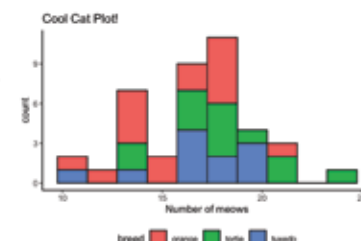
`geom_*(position = ...`
`position_jitter(width = 0.2)`
 • For bars e.g. `geom_bar()`
 "dodge"  "stack" 

Theming plots

```
ggplot(data = cat_data,
aes(x = meow_count,
fill = breed)) +
geom_histogram(bins = 10, Label plot
colour = "black") +
labs(title = "Cool Cat Plot!",
x = "Number of meows") +
theme_classic(base_size = 10) +
theme(legend.position = "bottom") ↑
Use a pre-set theme
```

Style further with `theme()` function

Remember you can set aesthetics outside `aes()` to style your plot!



Faceting

Faceting makes **multiple individual plots** based on the value of a variable:
 + `facet_wrap(~variable)` or even two: + `facet_grid(var1 ~ var2)`

Combining multiple plots

Load the package:
`library(patchwork)`

plot1 plot2



plot1 / plot2 plot1 + plot2

Add `plot_annotation(tag_levels = "A")`
 to label plots with A, B, C etc